

256 bit (16 x 16 or 32 x 8) SERIAL MICROWIRE EEPROM

NOT FOR NEW DESIGN

- 1 MILLION ERASE/WRITE CYCLES, with 40 YEARS DATA RETENTION
- DUAL ORGANIZATION: 16 x 16 or 32 x 8
- BYTE/WORD and ENTIRE MEMORY PROGRAMMING INSTRUCTIONS
- SELF-TIMED PROGRAMMING CYCLE with AUTO-ERASE
- READY/BUSY SIGNAL DURING PROGRAMMING
- SINGLE 5V $\pm 10\%$ SUPPLY VOLTAGE
- SEQUENTIAL READ OPERATION
- 5ms TYPICAL PROGRAMMING TIME
- ENHANCED ESD/LATCH UP PERFORMANCES for "C" VERSION
- **ST93C06 and ST93C06C are replaced by the M93C06**

DESCRIPTION

The ST93C06 and ST93C06C are 256 bit Electrically Erasable Programmable Memory (EEPROM) fabricated with SGS-THOMSON's High Endurance Single Polysilicon CMOS technology. In the text the two products are referred to as ST93C06.

The memory is divided into either 32 x 8 bit bytes or 16 x 16 bit words. The organization may be selected by a signal applied on the ORG input.

The memory is accessed through a serial input (D) and by a set of instructions which includes Read a byte/word, Write a byte/word, Erase a byte/word, Erase All and Write All. A Read instruction loads the address of the first byte/word to be read into an internal address pointer.

Table 1. Signal Names

S	Chip Select Input
D	Serial Data Input
Q	Serial Data Output
C	Serial Clock
ORG	Organisation Select
V _{CC}	Supply Voltage
V _{SS}	Ground

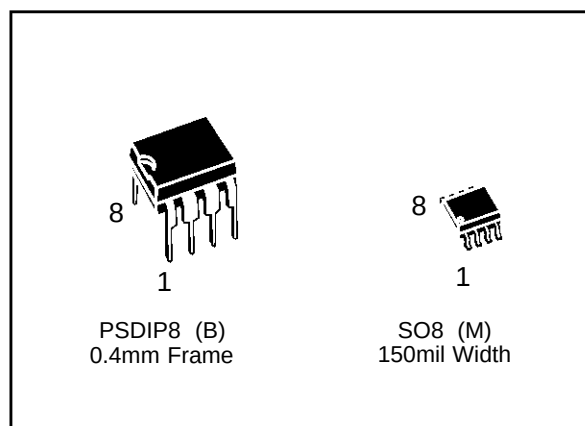


Figure 1. Logic Diagram

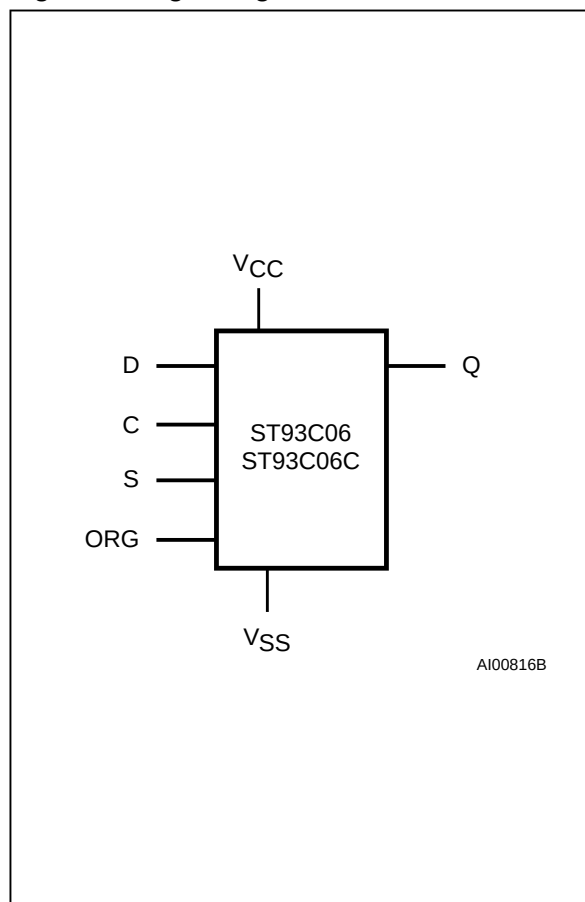


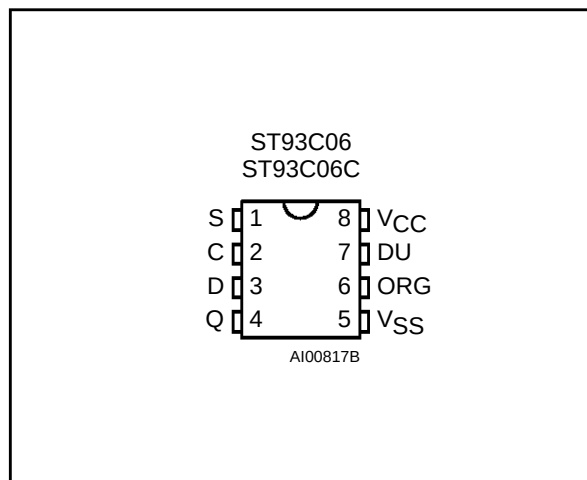
Figure 2A. DIP Pin Connections

Warning: DU = Don't Use

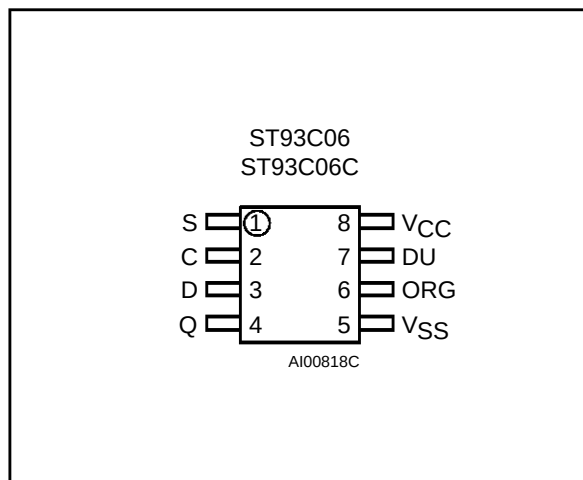
Figure 2B. SO Pin Connections

Warning: DU = Don't Use

Table 2. Absolute Maximum Ratings ⁽¹⁾

Symbol	Parameter	Value	Unit
T _A	Ambient Operating Temperature	−40 to 125	°C
T _{STG}	Storage Temperature	−65 to 150	°C
T _{LEAD}	Lead Temperature, Soldering (SO8 package) 40 sec (PSDIP8 package) 10 sec	215 260	°C
V _{IO}	Input or Output Voltages (Q = V _{OH} or Hi-Z)	−0.3 to V _{CC} +0.5	V
V _{CC}	Supply Voltage	−0.3 to 6.5	V
V _{ESD}	Electrostatic Discharge Voltage (Human Body model) ⁽²⁾ ST93C06 ST93C06C	2000 4000	V
	Electrostatic Discharge Voltage (Machine model) ⁽³⁾ ST93C06 ST93C06C	500 500	V

Notes: 1. Except for the rating "Operating Temperature Range", stresses above those listed in the Table "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only and operation of the device at these or any other conditions above those indicated in the Operating sections of this specification is not implied. Exposure to Absolute Maximum Rating conditions for extended periods may affect device reliability. Refer also to the SGS-THOMSON SURE Program and other relevant quality documents.

2. MIL-STD-883C, 3015.7 (100pF, 1500 Ω).

3. EIAJ IC-121 (Condition C) (200pF, 0 Ω).

DESCRIPTION (cont'd)

The data contained at this address is then clocked out serially. The address pointer is automatically incremented after the data is output and, if the Chip Select input (S) is held High, the ST93C06 can output a sequential stream of data bytes/words. In this way, the memory can be read as a data stream from 8 to 256 bits long, or continuously as the address counter automatically rolls over to '00' when the highest address is reached. Programming is internally self-timed (the external clock

signal on C input may be disconnected or left running after the start of a Write cycle) and does not require an erase cycle prior to the Write instruction. The Write instruction writes 8 or 16 bits at one time into one of the 32 bytes or 16 words. After the start of the programming cycle a Busy/Ready signal is available on the Data output (Q) when Chip Select (S) is driven High.

The design of the ST93C06 and the High Endurance CMOS technology used for its fabrication give an Erase/Write cycle Endurance of 1,000,000 cycles and a data retention of 40 years.

AC MEASUREMENT CONDITIONS

Input Rise and Fall Times	$\leq 20\text{ns}$
Input Pulse Voltages	0.4V to 2.4V
Input Timing Reference Voltages	1V to 2.0V
Output Timing Reference Voltages	0.8V to 2.0V

Note that Output Hi-Z is defined as the point where data is no longer driven.

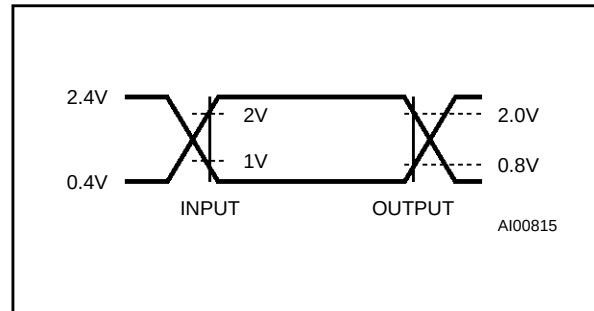
Figure 3. AC Testing Input Output Waveforms

Table 3. Capacitance⁽¹⁾
 ($T_A = 25^\circ\text{C}$, $f = 1\text{ MHz}$)

Symbol	Parameter	Test Condition	Min	Max	Unit
C_{IN}	Input Capacitance	$V_{IN} = 0V$		5	pF
C_{OUT}	Output Capacitance	$V_{OUT} = 0V$		5	pF

Note: 1. Sampled only, not 100% tested.

Table 4. DC Characteristics
 ($T_A = 0$ to 70°C or -40 to 85°C ; $V_{CC} = 5V \pm 10\%$)

Symbol	Parameter	Test Condition	Min	Max	Unit
I_{LI}	Input Leakage Current	$0V \leq V_{IN} \leq V_{CC}$		± 2.5	μA
I_{LO}	Output Leakage Current	$0V \leq V_{OUT} \leq V_{CC}$, Q in Hi-Z		± 2.5	μA
I_{CC}	Supply Current (TTL Inputs)	$S = V_{IH}$, $f = 1\text{ MHz}$		3	mA
	Supply Current (CMOS Inputs)	$S = V_{IH}$, $f = 1\text{ MHz}$		2	mA
I_{CC1}	Supply Current (Standby)	$S = V_{SS}$, $C = V_{SS}$, ORG = V_{SS} or V_{CC}		50	μA
V_{IL}	Input Low Voltage (D, C, S)		-0.3	0.8	V
V_{IH}	Input High Voltage (D, C, S)		2	$V_{CC} + 1$	V
V_{OL}	Output Low Voltage	$I_{OL} = 2.1\text{mA}$		0.4	V
		$I_{OL} = 10\mu A$		0.2	V
V_{OH}	Output High Voltage	$I_{OH} = -400\mu A$	2.4		V
		$I_{OH} = -10\mu A$	$V_{CC} - 0.2$		V

Table 5. AC Characteristics(T_A = 0 to 70°C or -40 to 85°C; V_{CC} = 5V ± 10%)

Symbol	Alt	Parameter	Test Condition	Min	Max	Unit
t _{SHCH}	t _{CSS}	Chip Select High to Clock High		50		ns
t _{CLSH}	t _{SKS}	Clock Low to Chip Select High		100		ns
t _{DVCH}	t _{DIS}	Input Valid to Clock High		100		ns
t _{CHDX}	t _{DIH}	Clock High to Input Transition	Temp. Range: grade 1	100		ns
			Temp. Range: grades 3, 6	200		ns
t _{CHQL}	t _{PD0}	Clock High to Output Low			500	ns
t _{CHQV}	t _{PD1}	Clock High to Output Valid			500	ns
t _{CLSL}	t _{CSH}	Clock Low to Chip Select Low		0		ns
t _{SLCH}		Chip Select Low to Clock High		250		ns
t _{SLSH}	t _{CS}	Chip Select Low to Chip Select High	Note 1	250		ns
t _{SHQV}	t _{SV}	Chip Select High to Output Valid			500	ns
t _{SLQZ}	t _{DF}	Chip Select Low to Output Hi-Z	ST93C06		300	ns
			ST93C06C		200	ns
t _{CHCL}	t _{SKH}	Clock High to Clock Low	Note 2	250		ns
t _{CLCH}	t _{SKL}	Clock Low to Clock High	Note 2	250		ns
t _w	t _{WP}	Erase/Write Cycle time			10	ms
f _C	f _{SK}	Clock Frequency		0	1	MHz

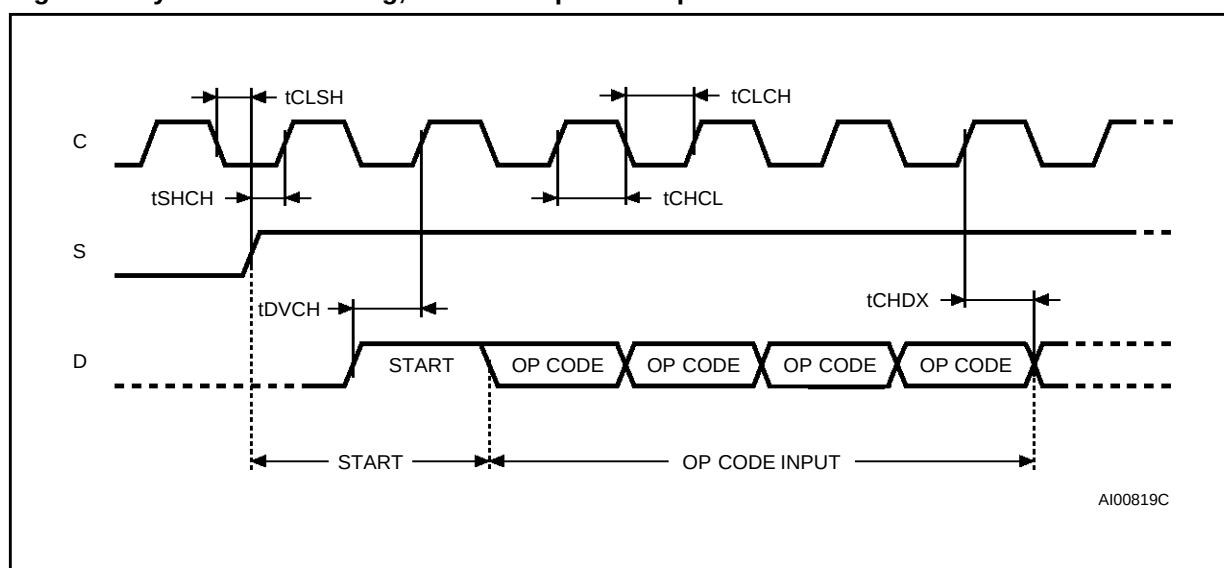
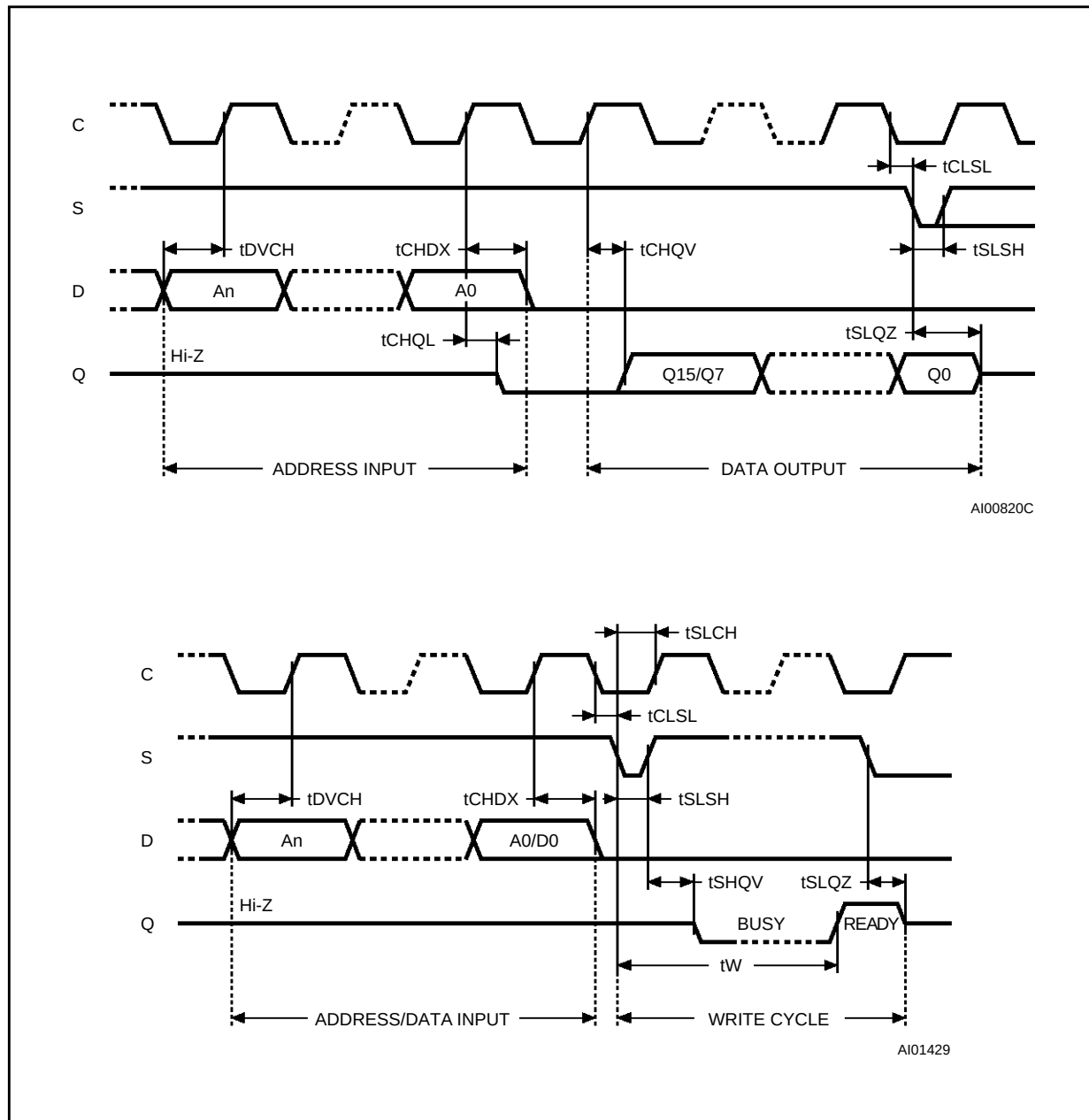
Notes: 1. Chip Select must be brought low for a minimum of 250 ns (t_{SLSH}) between consecutive instruction cycles.2. The Clock frequency specification calls for a minimum clock period of 1 µs, therefore the sum of the timings t_{CHCL} + t_{CLCH} must be greater or equal to 1 µs. For example, if t_{CHCL} is 250 ns, then t_{CLCH} must be at least 750 ns.**Figure 4. Synchronous Timing, Start and Op-Code Input**

Figure 5. Synchronous Timing, Read or Write

**DESCRIPTION (cont'd)**

The DU (Don't Use) pin does not affect the function of the memory and it is reserved for use by SGS-THOMSON during test sequences. The pin may be left unconnected or may be connected to V_{CC} or V_{SS} . Direct connection of DU to V_{SS} is recommended for the lowest standby power consumption.

MEMORY ORGANIZATION

The ST93C06 is organized as 32 bytes x 8 bits or 16 words x 16 bits. If the ORG input is left unconnected (or connected to V_{CC}) the x16 organization is selected, when ORG is connected to Ground (V_{SS}) the x8 organization is selected. When the ST93C06 is in standby mode, the ORG input should be unconnected or set to either V_{SS} or V_{CC} in order to achieve the minimum power consumption. Any voltage between V_{SS} and V_{CC} applied to ORG may increase the standby current value.

POWER-ON DATA PROTECTION

In order to prevent data corruption and inadvertent write operations during power up, a Power On Reset (POR) circuit resets all internal programming circuitry and sets the device in the Write Disable mode. When V_{CC} reaches its functional value, the device is properly reset (in the Write Disable mode) and is ready to decode and execute an incoming instruction. A stable V_{CC} must be applied before any logic signal.

INSTRUCTIONS

The ST93C06 has seven instructions, as shown in Table 6. The op-codes of the instructions are made up of 4 bits: some instructions use only the first two bits, others use all four bits to define the op-code. The op-code is followed by an address for the byte/word which is four bits long for the x16 organization or five bits long for the x8 organization.

Each instruction is preceded by the rising edge of the signal applied on the S input (assuming that clock C and data input D are low), followed by a first clock pulse which is ignored by the ST93C06 (optional clock pulse for the ST93C06C). The data input D is then sampled upon the following rising edges of the clock C until a '1' is sampled and decoded by the ST93C06 as a Start bit. Even though the first clock pulse is ignored, it recommended to pull low the data input D during this first clock pulse in order to keep the timing upwardly compatible with other ST93Cxx devices.

The ST93C06 is fabricated in CMOS technology and is therefore able to run from zero Hz (static input signals) up to the maximum ratings (specified in Table 5).

Read

The Read instruction (READ) outputs serial data on the Data Output (Q). When a READ instruction is received, the instruction and address are decoded and the data from the memory is transferred into an output shift register. A dummy '0' bit is output first followed by the 8 bit byte or the 16 bit word with the MSB first. Output data changes are triggered by the Low to High transition of the Clock (C). The ST93C06 will automatically increment the address and will clock out the next byte/word as long as the Chip Select input (S) is held High. In this case the dummy '0' bit is NOT output between bytes/words and a continuous stream of data can be read.

Erase/Write Enable and Disable

The Erase/Write Enable instruction (EWEN) authorizes the following Erase/Write instructions to be executed, the Erase/Write Disable instruction (EWDS) disables the execution of the following Erase/Write instructions. When power is first applied, the ST93C06 enters the Disable mode. When the Erase/Write Enable instruction (EWEN) is executed, Write instructions remain enabled until an Erase/Write Disable instruction (EWDS) is executed or if the Power-on reset circuit becomes active due to a reduced V_{CC} . To protect the memory contents from accidental corruption, it is advisable to issue the EWDS instruction after every write cycle. The READ instruction is not affected by the EWEN or EWDS instructions.

Erase

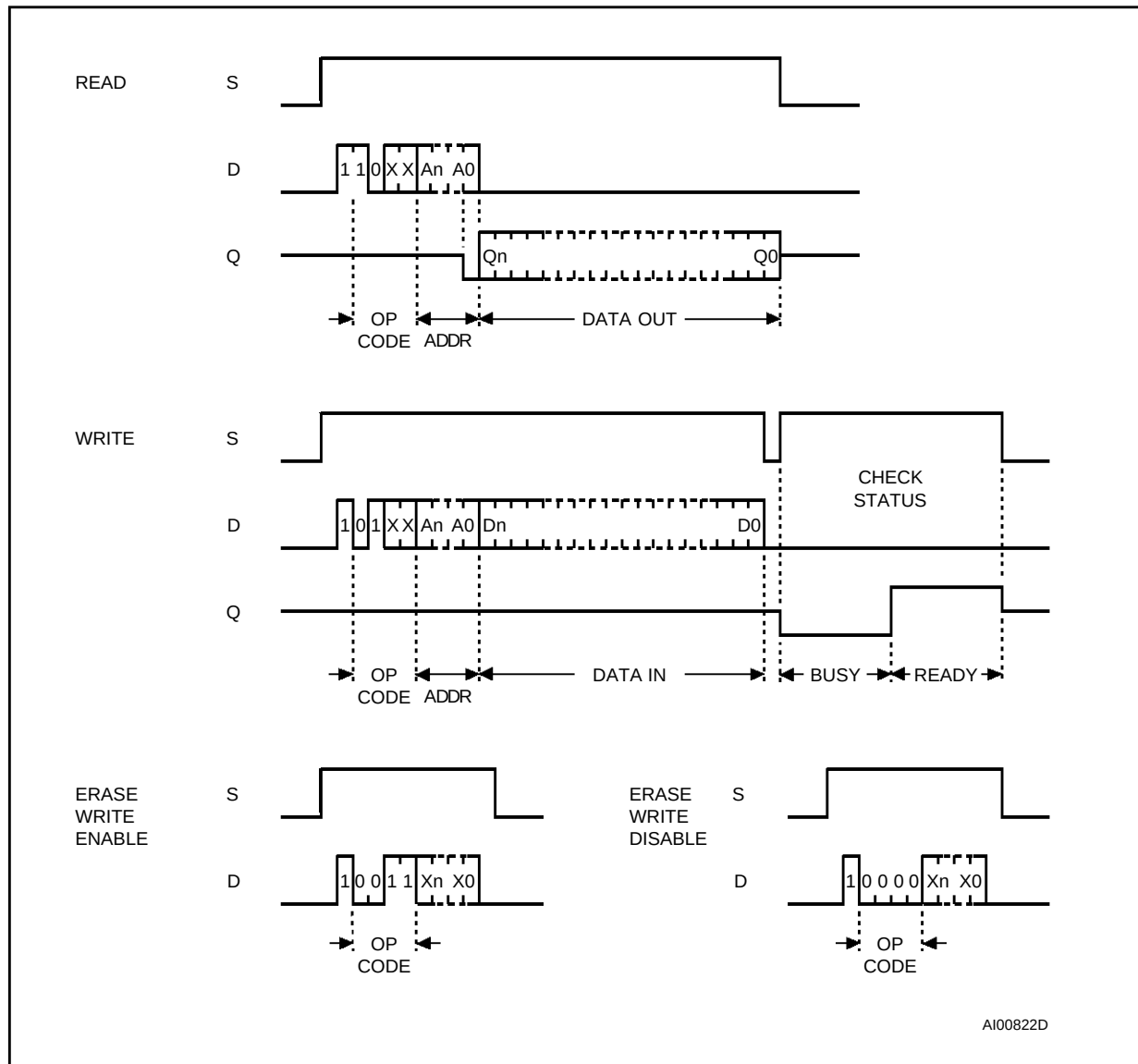
The Erase instruction (ERASE) programs the addressed memory byte or word bits to '1'. Once the address is correctly decoded, the falling edge of the Chip Select input (S) triggers a self-timed erase cycle.

Table 6. Instruction Set

Instruction	Description	Op-Code	x8 Org Address (ORG = 0)	Data	x16 Org Address (ORG = 1)	Data
READ	Read Data from Memory	10XX	A4-A0	Q7-Q0	A3-A0	Q15-Q0
WRITE	Write Data to Memory	01XX	A4-A0	D7-D0	A3-A0	D15-D0
EWEN	Erase/Write Enable	0011	XXXXX		XXXX	
EWDS	Erase/Write Disable	0000	XXXXX		XXXX	
ERASE	Erase Byte or Word	11XX	A4-A0		A3-A0	
ERAL	Erase All Memory	0010	XXXXX		XXXX	
WRAL	Write All Memory with same Data	0001	XXXXX	D7-D0	XXXX	D15-D0

Note: X = don't care bit.

Figure 6. READ, WRITE, EWEN, EWDS Sequences



Notes: 1. A_n: n = 3 for x16 org. and 4 for x8 org.
 2. X_n: n = 3 for x16 org. and 4 for x8 org.

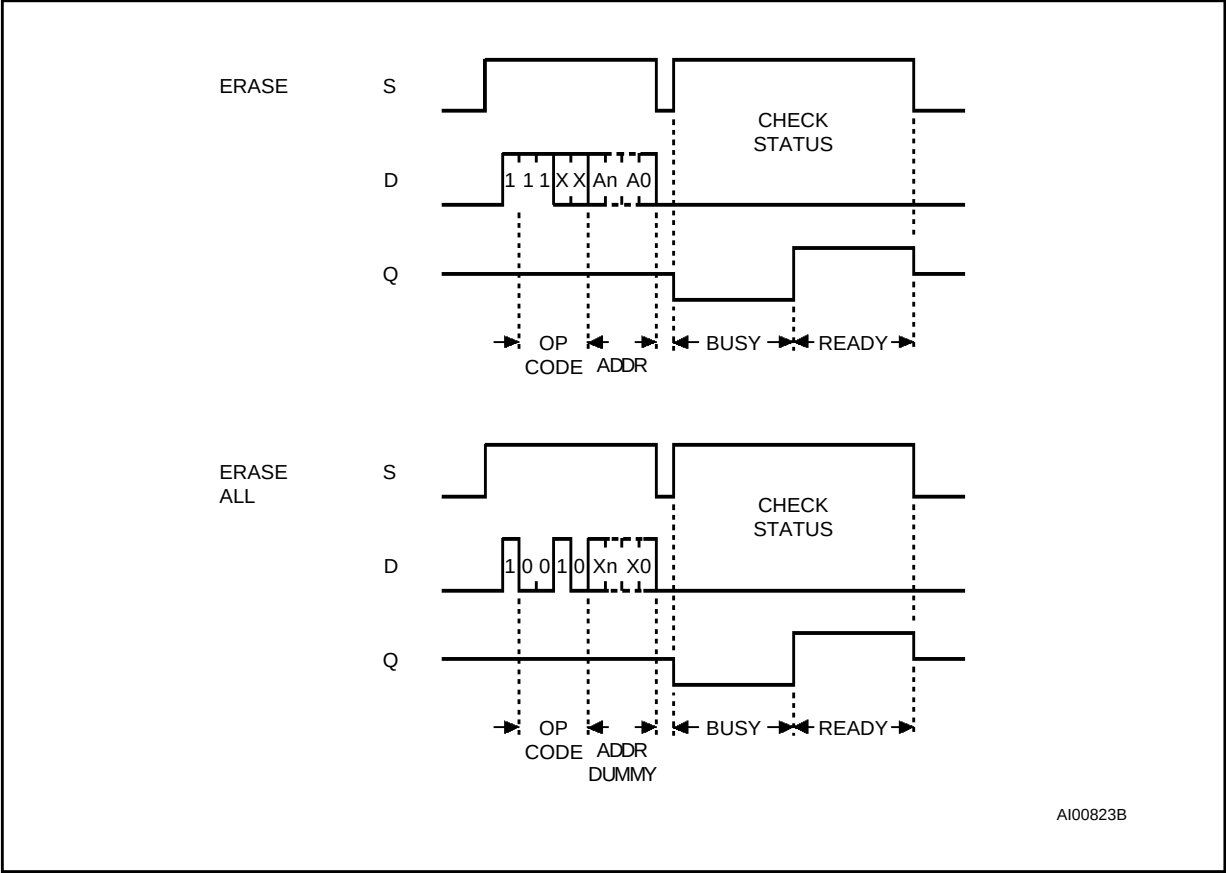
If the ST93C06 is still performing the erase cycle, the Busy signal (Q = 0) will be returned if S is driven high, and the ST93C06 will ignore any data on the bus. When the erase cycle is completed, the Ready signal (Q = 1) will indicate (if S is driven high) that the ST93C06 is ready to receive a new instruction.

Write

The Write instruction (WRITE) is followed by the address and the 8 or 16 data bits to be written. Data input is sampled on the Low to High transition of the clock. After the last data bit has been sampled, Chip Select (S) must be brought Low before the next rising edge of the clock (C) in order to start the

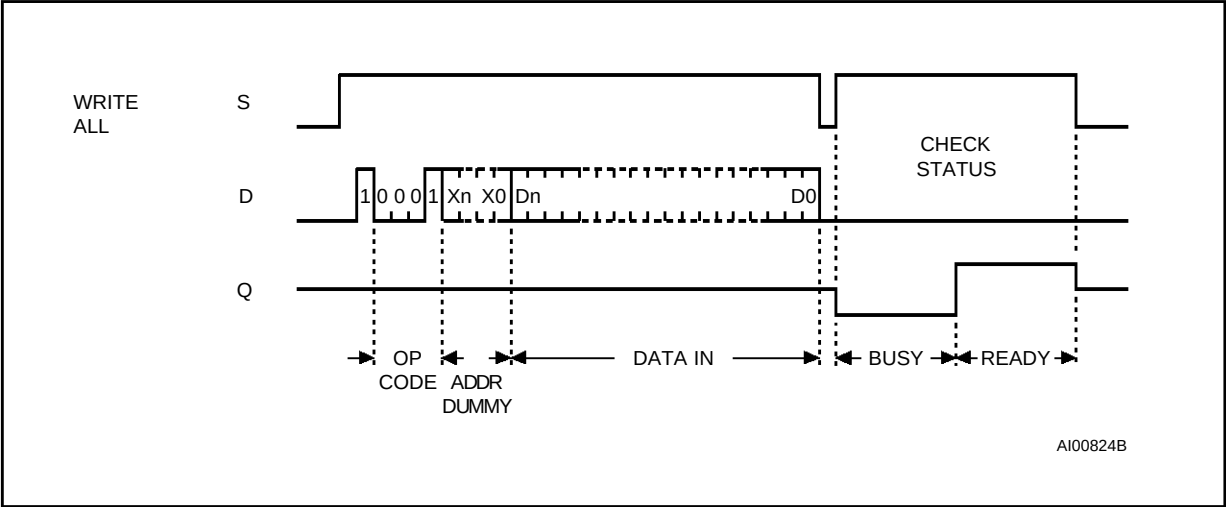
self-timed programming cycle. If the ST93C06 is still performing the write cycle, the Busy signal (Q = 0) will be returned if S is driven high, and the ST93C06 will ignore any data on the bus. When the write cycle is completed, the Ready signal (Q = 1) will indicate (if S is driven high) that the ST93C06 is ready to receive a new instruction. Programming is internally self-timed (the external clock signal on C input may be disconnected or left running after the start of a programming cycle) and does not require an Erase instruction prior to the Write instruction (The Write instruction includes an automatic erase cycle before programming data).

Figure 7. ERASE, ERAL Sequences



Notes: 1. A_n: n = 3 for x16 org. and 4 for x8 org.
2. X_n: n = 3 for x16 org. and 4 for x8 org.

Figure 8. WRAL Sequence



Note: 1 X_n: n = 3 for x16 org. and 4 for x8 org.

Erase All

The Erase All instruction (ERAL) erases the whole memory (all memory bits are set to '1'). A dummy address is input during the instruction transfer and the erase is made in the same way as the ERASE instruction. If the ST93C06 is still performing the erase cycle, the Busy signal ($Q = 0$) will be returned if S is driven high, and the ST93C06 will ignore any data on the bus. When the erase cycle is completed, the Ready signal ($Q = 1$) will indicate (if S is driven high) that the ST93C06 is ready to receive a new instruction.

Write All

For correct operation, an ERAL instruction should be executed before the WRAL instruction: the WRAL instruction DOES NOT perform an automatic erase before writing. The Write All instruction (WRAL) writes the Data Input byte or word to all the addresses of the memory. If the ST93C06 is still performing the write cycle, the Busy signal ($Q = 0$) will be returned if S is driven high, and the ST93C06 will ignore any data on the bus. When the write cycle is completed, the Ready signal ($Q = 1$) will indicate (if S is driven high) that the ST93C06 is ready to receive a new instruction.

READY/BUSY Status

During every programming cycle (after a WRITE, ERASE, WRAL or ERAL instruction) the Data Output (Q) indicates the Ready/Busy status of the

memory when the Chip Select (S) is driven High. Once the ST93C06 is Ready, the Ready/Busy status is available on the Data Output (Q) until a new start bit is decoded or the Chip Select (S) is brought Low.

COMMON I/O OPERATION

The Data Output (Q) and Data Input (D) signals can be connected together, through a current limiting resistor, to form a common, one wire data bus. Some precautions must be taken when operating the memory with this connection, mostly to prevent a short circuit between the last entered address bit (A0) and the first data bit output by Q. The reader may also refer to the SGS-THOMSON application note "MICROWIRE EEPROM Common I/O Operation".

DIFFERENCES BETWEEN ST93C06 AND ST93C06C

Each instruction of the ST93C06 requires an Additional Dummy clock pulse after the rising edge of the Chip Select input (S) and before the START bit, see Figure 9. When replacing the ST93C06 with the ST93C06C in an application, it must be checked that this Dummy Clock cycle DOES NOT HAPPEN when D = 1: if it is so, this clock pulse will latch an information which is decoded by the ST93C06C as a START bit (see Figure 10) and the following bits will be decoded with a shift of one bit.

Figure 9. ST93C06 Timing

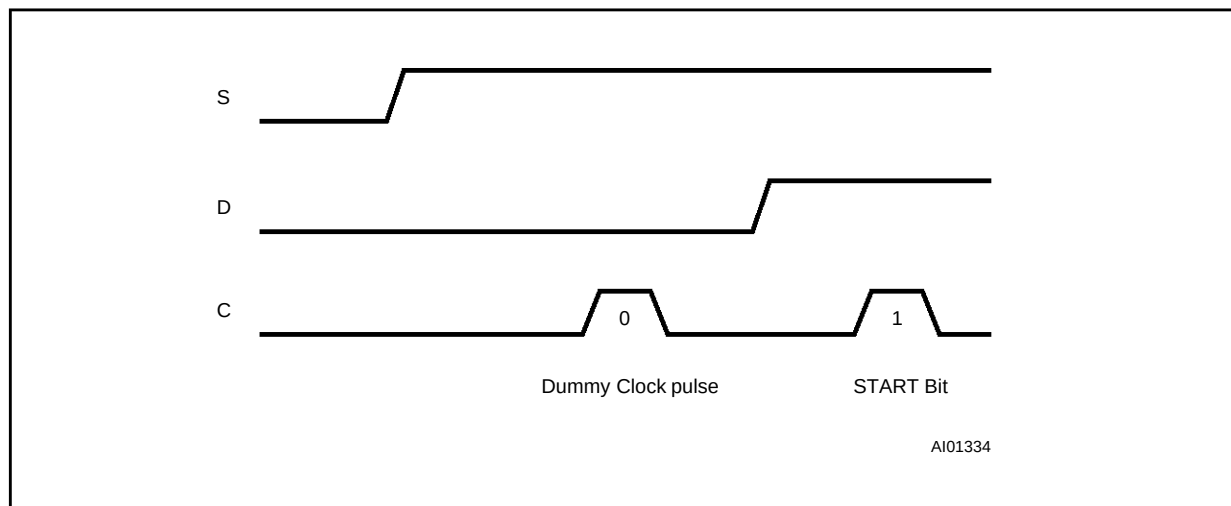


Figure 10. Comparative Timings

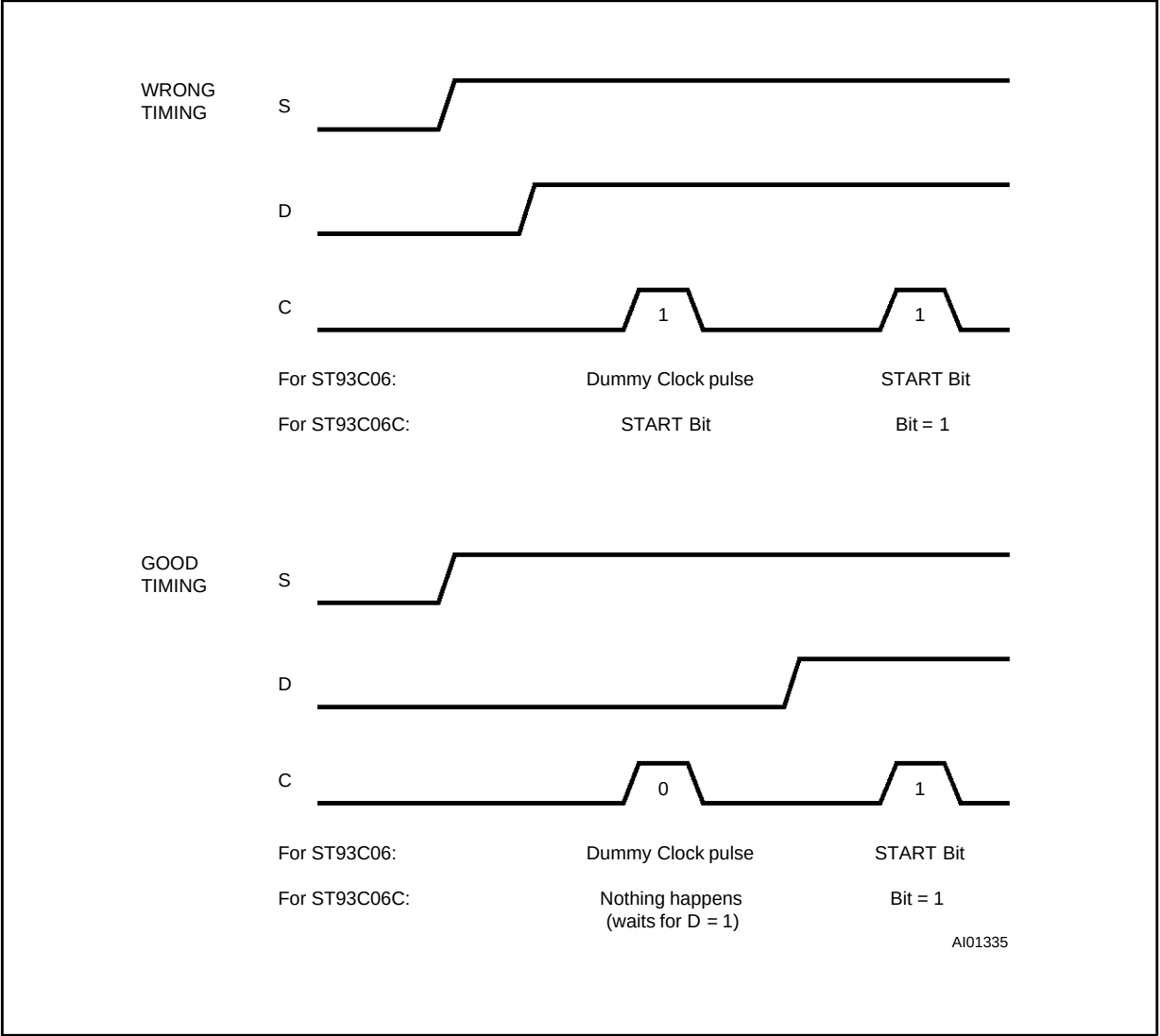
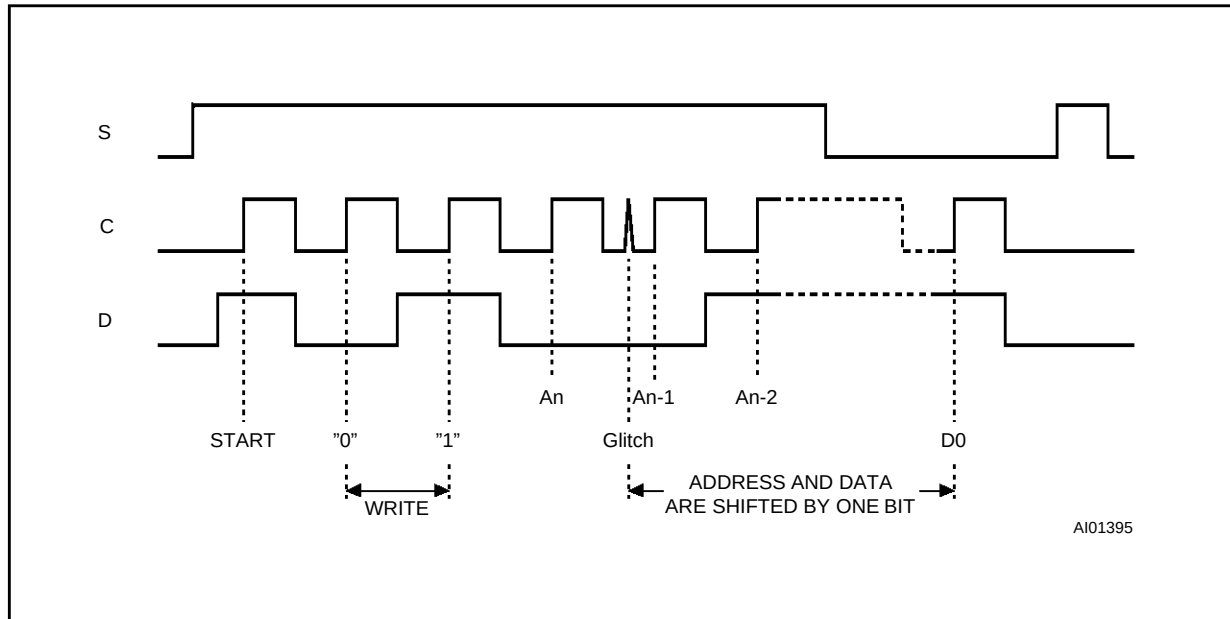


Figure 11. WRITE Sequence with One Clock Glitch



DIFFERENCES BETWEEN ST93C06 AND ST93C06C (cont'd)

The ST93C06C is an enhanced version of the ST93C06A and offers the following extra features:

- Enhanced ESD voltage
- Functional security filtering glitches on the clock input (C).

Refer to Table 2 (Absolute Maximum Ratings) for more about ESD limits. The following description will detail the Clock pulses counter (available only on the ST93C06C).

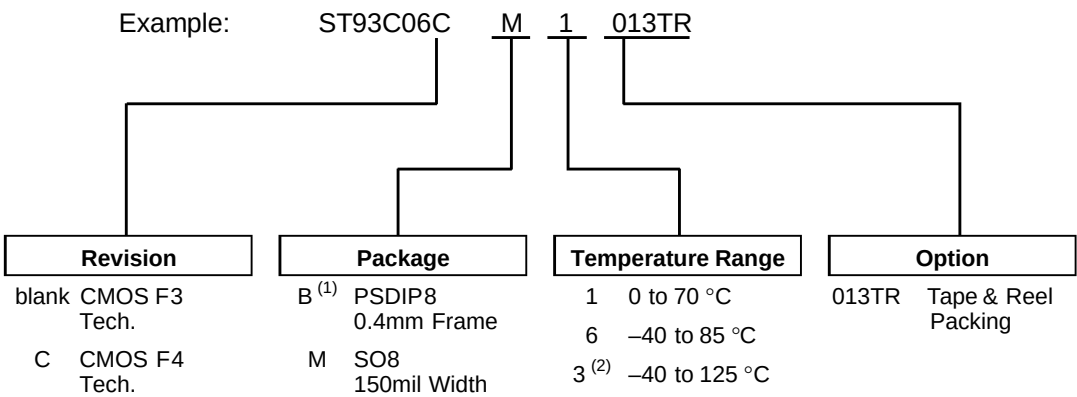
In a normal environment, the ST93C06 is expected to receive the exact amount of data on the D input, that is the exact amount of clock pulses on the C input.

In a noisy environment, the amount of pulses received (on the clock input C) may be greater than the clock pulses delivered by the Master (Microcontroller) driving the ST93C06C. In such a case, a part of the instruction is delayed by one bit (see Figure 11), and it may induce an erroneous write of data at a wrong address.

The ST93C46C has an on-board counter which counts the clock pulses from the Start bit until the falling edge of the Chip Select signal. For the WRITE instructions, the number of clock pulses incoming to the counter must be exactly 18 (with the Organisation by 8) from the Start bit to the falling edge of Chip Select signal (1 Start bit + 2 bits of Op-code + 7 bits of Address + 8 bits of Data = 18): if so, the ST93C06C executes the WRITE instruction; if the number of clock pulses is not equal to 18, the instruction will not be executed (and data will not be corrupted).

In the same way, when the Organisation by 16 is selected, the number of clock pulses incoming to the counter must be exactly 25 (1 Start bit + 2 bits of Op-code + 6 bits of Address + 16 bits of Data = 25) from the Start bit to the falling edge of Chip Select signal: if so, the ST93C06C executes the WRITE instruction; if the number of clock pulses is not equal to 25, the instruction will not be executed (and data will not be corrupted). The clock pulse counter is active only on ERASE and WRITE instructions (WRITE, ERASE, ERAL, WRALL).

ORDERING INFORMATION SCHEME



Notes: 1. ST93C06CB1 is available with 0.25mm lead Frame only.
2. Temperature range on special request only.

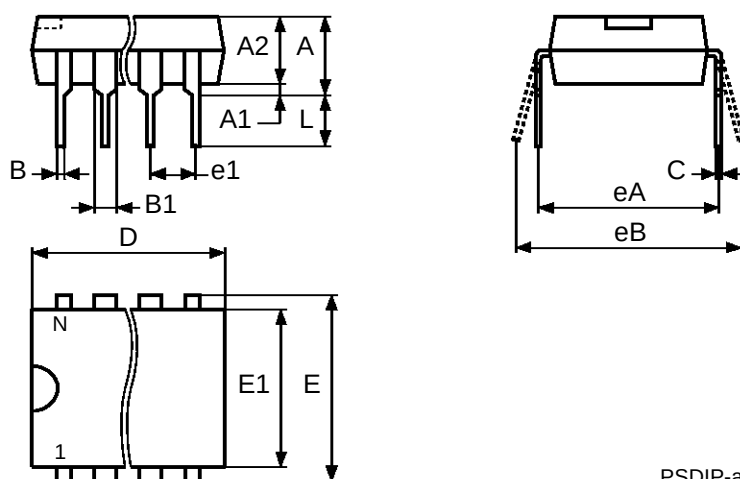
Devices are shipped from the factory with the memory content set at all "1's" (FFFFh for x16, FFh for x8).

For a list of available options (Package, etc...) or for further information on any aspect of this device, please contact the SGS-THOMSON Sales Office nearest to you.

PSDIP8 - 8 pin Plastic Skinny DIP, 0.4mm lead frame

Symb	mm			inches		
	Typ	Min	Max	Typ	Min	Max
A			4.80			0.189
A1		0.70	–		0.028	–
A2		3.10	3.60		0.122	0.142
B		0.38	0.58		0.015	0.023
B1		1.15	1.65		0.045	0.065
C		0.38	0.52		0.015	0.020
D		9.20	9.90		0.362	0.390
E	7.62	–	–	0.300	–	–
E1		6.30	7.10		0.248	0.280
e1	2.54	–	–	0.100	–	–
eA		8.40	–		0.331	–
eB			9.20			0.362
L		3.00	3.80		0.118	0.150
N		8			8	

PSDIP8



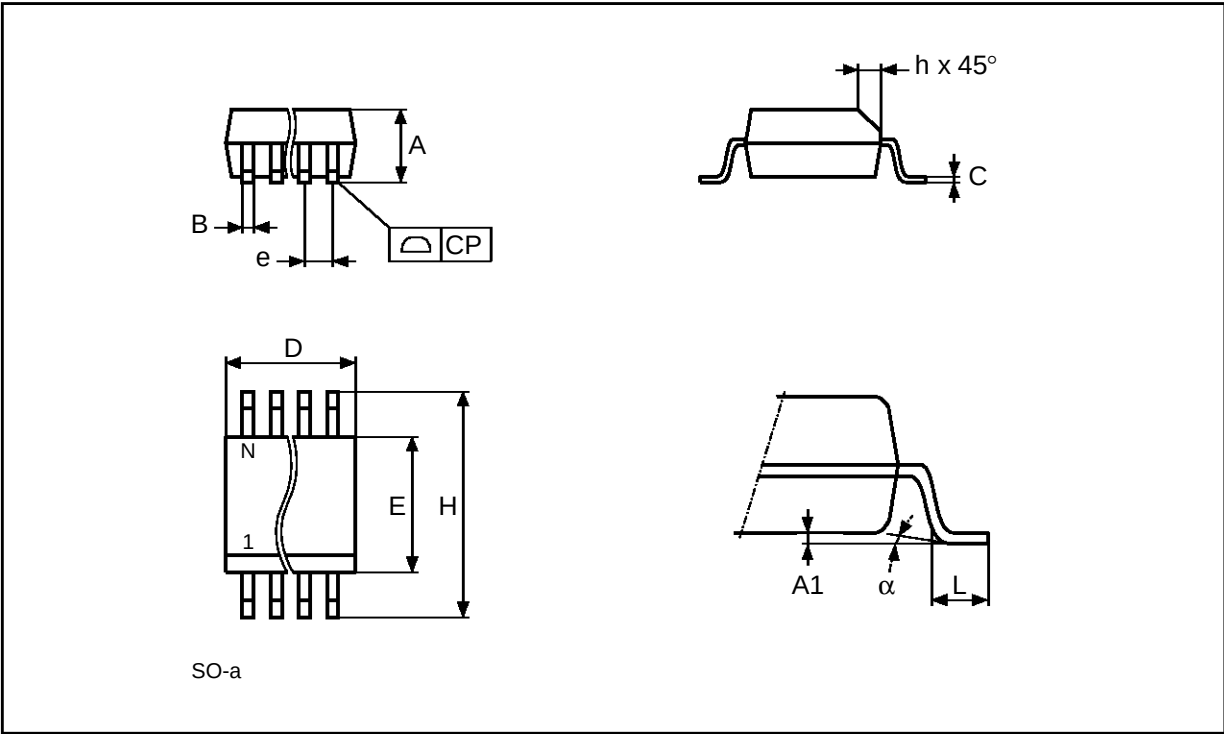
PSDIP-a

Drawing is not to scale

SO8 - 8 lead Plastic Small Outline, 150 mils body width

Symb	mm			inches		
	Typ	Min	Max	Typ	Min	Max
A		1.35	1.75		0.053	0.069
A1		0.10	0.25		0.004	0.010
B		0.33	0.51		0.013	0.020
C		0.19	0.25		0.007	0.010
D		4.80	5.00		0.189	0.197
E		3.80	4.00		0.150	0.157
e	1.27	—	—	0.050	—	—
H		5.80	6.20		0.228	0.244
h		0.25	0.50		0.010	0.020
L		0.40	0.90		0.016	0.035
α		0°	8°		0°	8°
N	8			8		
CP			0.10			0.004

SO8



Drawing is not to scale

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